

LYRA: GENERATIVE 3D SCENE RECONSTRUCTION VIA VIDEO DIFFUSION MODEL SELF-DISTILLATION

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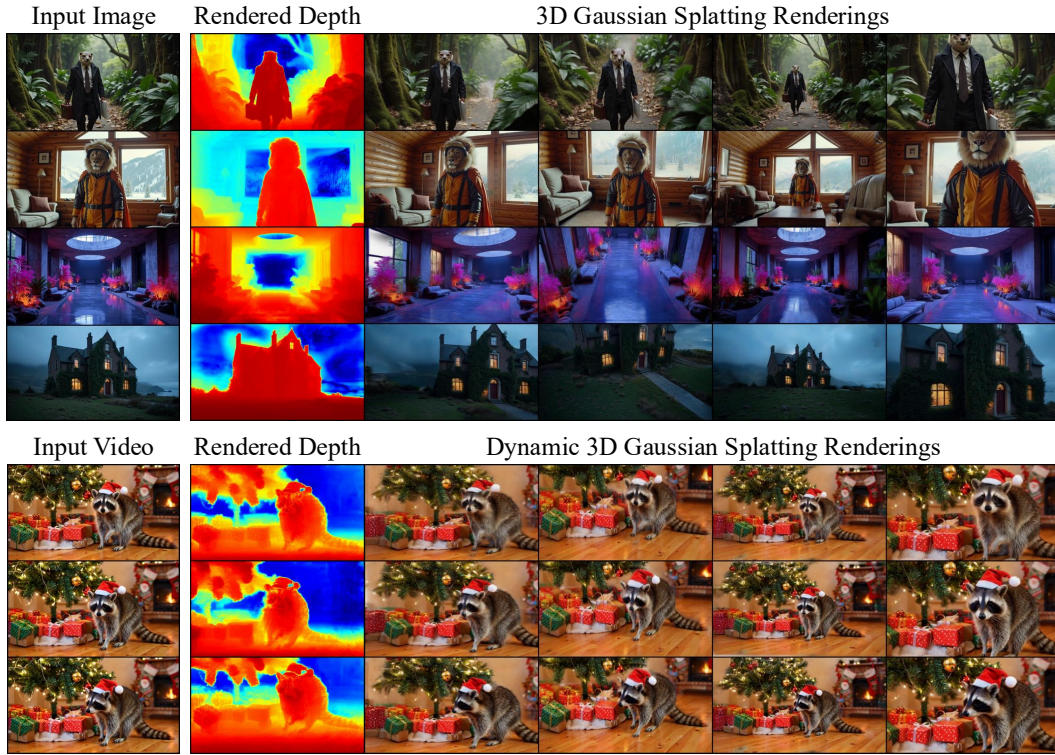


Figure 1: **Feed-Forward 3D and 4D Scene Generation.** From a single image (top), *Lyra* infers a 3D Gaussian Splatting (3DGS) representation in a single feed-forward pass by self-distilling a video diffusion model, without any real-world multi-view data. Given a video input (bottom), *Lyra* infers a dynamic 3DGS that affords interactive control over both time (rows) and viewpoint (columns).

ABSTRACT

The capacity to generate virtual environments underpins applications that range from gaming to the physical-AI domains of robotics, autonomous driving, and industrial AI. Prevailing learning-based 3D reconstruction methods, however, are contingent on captured real-world multi-view data, a resource that is neither uniformly available nor cheaply scaled. Video diffusion models have recently demonstrated a striking capacity for visual imagination, yet their intrinsically 2D output precludes the direct use required by simulation, where a robot must navigate and physically interact with its surroundings. In this paper, we resolve this gap through a self-distillation framework that distills the implicit 3D knowledge embedded in video diffusion models into an explicit 3D Gaussian Splatting (3DGS) representation, thereby eliminating the need for multi-view training data. Concretely, we augment the standard RGB decoder with a parallel 3DGS decoder that is supervised directly by the output of the RGB decoder, so that the 3DGS decoder is trained entirely from synthetic data generated by the video diffusion model. At inference time, the model synthesizes an explicit 3D scene from either a text prompt or a single image and renders it in real time, and the same framework extends naturally

to dynamic 3D scene generation from a monocular input video. Across both static and dynamic settings, our framework attains state-of-the-art 3D scene generation.

1 INTRODUCTION

Creating high-quality 3D environments at scale is a long-standing challenge in computer vision and graphics, one that underpins applications across film-making, VR/AR, and physical AI with closed-loop simulation. What these applications demand in common is an explicit 3D representation that simultaneously supports real-time rendering, physical interaction, and consistent multi-view synthesis.

Recent advances in neural 3D reconstruction (Mildenhall et al., 2020; Kerbl et al., 2023; Moenne-Loccoz et al., 2024) recover precisely such representations from posed images. Their scalability, however, is fundamentally constrained by a reliance on accurate camera poses and high-quality imagery, and the difficulty compounds for dynamic scenes, which typically demand synchronized multi-camera rigs (Li et al., 2022). More fundamentally, reconstruction is confined to observed content and cannot extrapolate beyond the input views. Feed-forward reconstruction models (Zhang et al., 2024e; Ren et al., 2024c) were introduced to relax these constraints, yet they inherit a bottleneck of their own: the scarcity of diverse, large-scale 3D training data, which curtails out-of-domain generalization.

Video diffusion models (Cosmos, 2025; Wan, 2025) offer a compelling way out of this impasse. Trained on internet-scale video corpora, they attain high fidelity and broad generalization across diverse environments. Because they learn from real-world footage spanning a wide gamut of camera trajectories, from handheld motion to cinematic panning and drone flyovers, these models implicitly encode rich cues about the underlying 3D world without ever observing explicit multi-view supervision. Moreover, unlike reconstruction approaches (Charatan et al., 2024), generative models can synthesize plausible content beyond what the input frames reveal. The obstacle is that video diffusion models emit only 2D frames and carry no explicit 3D representation, which restricts their use in simulation and in downstream tasks that demand geometric consistency, long-term coherence, and physical interaction. Recent work (Wang et al., 2024e; Ren et al., 2025), however, establishes that video models can be endowed with explicit camera control using only modestly sized posed datasets. This ability to generate *posed* image sequences is pivotal: it converts a video model into a resource that can serve simultaneously as input and as supervision for 3D reconstruction.

Building on this insight, we unify the two paradigms of reconstruction and generation into a single formulation that we term *generative 3D scene reconstruction*. We introduce *Lyra*, a method that generates explicit 3D environments directly from the latent representations of a video diffusion model in a single forward pass. The core of *Lyra* is a self-distillation framework (Fig. 2) in which a 3D Gaussian Splatting (3DGS) decoder operates directly within the latent space of the video diffusion model. Given a single input image or video, we sample a camera trajectory as conditioning, denoise the resulting video latent, and decode it along two parallel branches: the standard RGB decoder synthesizes a video sequence, while our 3DGS decoder produces an explicit 3D representation. The two branches close a self-distillation loop in which the RGB branch (teacher) supervises the 3DGS branch (student). To broaden viewpoint coverage, we sample multiple camera trajectories, generate the corresponding video latents, and train the 3DGS decoder to fuse information across them while reconciling long-term and multi-view inconsistencies.

This generative reconstruction framework confers three decisive advantages. **(i)** It generates large-scale synthetic environments spanning diverse scenarios directly from video diffusion models, dispensing entirely with real-world multi-view captures. **(ii)** By operating in the latent space of the video model, it processes many views efficiently, sidestepping the heavy memory overhead that burdens pixel-space feed-forward reconstruction methods. **(iii)** Its explicit 3DGS output enforces geometric consistency and yields representations that are immediately usable in downstream tasks such as physical simulation. As a direct consequence, at inference time *Lyra* produces high-quality 3D Gaussian scenes from monocular input that render in real time *without any additional optimization or post-processing*.

We further extend the self-distillation framework to dynamic 4D generation from monocular video. Here the video model (teacher) supplies space-time supervision, while the student learns time-conditioned 3DGS representations that enable novel-view synthesis of dynamic scenes.

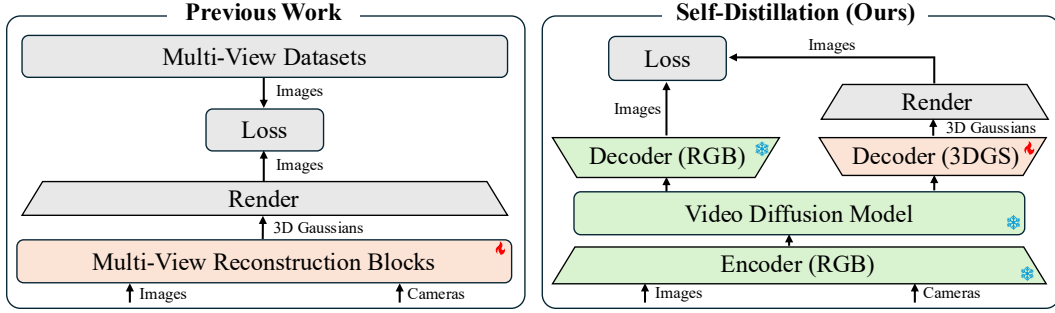


Figure 2: **Self-distillation framework of *Lyra***. Prior work (left) (Szymanowicz et al., 2025b) trains multi-view reconstruction blocks on real-world datasets of limited diversity (Zhou et al., 2018; Ling et al., 2024b). We instead propose a self-distillation framework (right) for generative 3D scene reconstruction, in which the RGB-decoder output (teacher) of a pre-trained camera-controlled video diffusion model supervises the renderings of the 3DGS decoder (student). Because the video model is pre-trained on diverse 2D video, it supplies correspondingly diverse multi-view supervision.

In summary, this work makes the following contributions:

- We introduce a self-distillation framework that trains a 3DGS student decoder using a pre-trained camera-controlled video diffusion model as the teacher, removing the dependence on captured multi-view real-world data entirely.
- We show that the same framework extends to dynamic scenes, enabling 4D reconstruction from a monocular video, a setting unexplored by prior feed-forward methods.
- We demonstrate that the resulting model generalizes across diverse scenes, establishing state-of-the-art results in both single-image 3D scene generation and single-video 4D scene generation.

2 RELATED WORK

Multi-view image generation. Early work on multi-view image generation (Watson et al., 2023; Liu et al., 2023; Shi et al., 2024; Wang & Shi, 2023) concentrated on object-centric scenes devoid of background, and subsequent efforts (Sargent et al., 2024; Tang et al., 2023b; Schneider et al., 2025) scaled multi-view image generators up to full scenes. CAT3D (Gao et al., 2024b) repurposes a pre-trained image diffusion model to generate multi-view images from any generated or real input image; a second stage then reconstructs the generated views into an explicit representation such as NeRF (Mildenhall et al., 2020) or 3DGS (Kerbl et al., 2023). This paradigm has further been extended to dynamic 3D scenes rendered from multiple viewpoints via space-time diffusion models (Watson et al., 2024; Wu et al., 2025b; Kuang et al., 2024; Wang et al., 2025a). Impressive as these results are, grounding the generations in an explicit 3D representation hinges on an expensive per-scene optimization stage whose cost is not amortized across scenes. Rather than synthesizing multi-view images and subsequently reconstructing them by optimization, we generate a 3D scene *directly* from text or a single image.

Camera-conditioned video generation. Fine-tuning video diffusion models for explicit 3D camera control has advanced rapidly. MotionCtrl (Wang et al., 2024e) pioneered this direction by conditioning pre-trained video models on camera poses. Follow-up work (He et al., 2024a; Xu et al., 2024c; Bahmani et al., 2025a;b) encodes cameras as Plücker coordinates for pixel-wise conditioning, while a complementary line (Hu et al., 2024; Hou et al., 2024; Zhang et al., 2024b) pursues training-free camera control of video diffusion models. More recently, SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a) achieved compelling camera-controlled video synthesis from large-scale synthetic training data. Compelling as they are, these methods produce outputs that remain ungrounded in 3D. Our work is orthogonal and complementary: we address feed-forward 3D generation from camera-controlled video models, so that novel viewpoints can be rendered *after* generation. Concretely, we build upon GEN3C (Ren et al., 2025), a recent camera-controlled video diffusion model, and employ it as the teacher within our self-distillation framework for 3D reconstruction.

Feed-forward 3D models. Early feed-forward 3D reconstruction targeted object-centric scenes without background, predicting a NeRF (Hong et al., 2024; Li et al., 2024b) or 3DGS (Tang et al., 2024a; Szymanowicz et al., 2024) directly from text or an image. More recent methods (Charatan et al., 2024; Zhang et al., 2024e; Szymanowicz et al., 2025a; Ren et al., 2024c) address scene-level reconstruction,

typically regressing per-pixel 3D Gaussians from one or more input images. Most such scene-level methods, however, remain tethered to their training distribution, e.g., RealEstate10K (Zhou et al., 2018), and generalize poorly to generated scenes. A further line of work extends feed-forward reconstruction toward scene *generation*: Bolt3D (Szymanowicz et al., 2025b) trains a pointmap (Wang et al., 2024b) autoencoder to jointly generate pointmaps and multi-view images for feed-forward reconstruction, and, most closely related to us, Wonderland (Liang et al., 2025a) uses a camera-controlled video model to produce 3D Gaussians with a feed-forward network. In contrast, we treat a camera-controlled video model as a teacher within a self-distillation framework and thereby train our student 3D model *without any real-world multi-view data*. Crucially, we further show that this framework admits a simple extension to feed-forward 4D scene generation, a task that has remained unexplored to date.

3 SELF-DISTILLATION USING VIDEO DIFFUSION MODELS

The central idea of our approach is to distill the implicit 3D knowledge embedded in video diffusion models into an explicit 3DGS decoder that produces high-quality 3D representations. We realize this idea through a teacher–student framework in which the video diffusion model (teacher) generates RGB videos that supervise a 3DGS decoder (student) operating within the same latent space, as illustrated in Fig. 2. This section first details the video diffusion model backbone (Sec. 3.1) and its role in self-distillation for the 3DGS decoder (Sec. 3.2). We then justify the design of the 3DGS decoder in Sec. 4 and specify the minimal modifications that adapt the pipeline to dynamic 3D scenes in Sec. 5.

3.1 BACKGROUND: CAMERA-CONTROLLED AND 3D-CONSISTENT VIDEO DIFFUSION

Because our 3DGS decoder is supervised exclusively by the camera-controlled video diffusion model, the 3D consistency of that model is a prerequisite for our method. We therefore build on GEN3C (Ren et al., 2025), a recent camera-conditioned video diffusion model, and summarize its key design choices below.

Spatiotemporal 3D cache. To sharpen both video consistency and camera-control precision, GEN3C constructs a spatiotemporal 3D cache $\{\mathbf{P}^{t,v}\}$ from the input image(s) or videos, where each $\mathbf{P}^{t,v}$ is a colored point cloud obtained by unprojecting the estimated depth (Wang et al., 2024a) of an RGB image captured from camera viewpoint v at time t . The cache is organized as an $L \times V$ array, in which L denotes the temporal length (number of frames) and V the number of camera views.

Rendering and structured guidance. To exploit this cache, GEN3C renders each point cloud from an arbitrary target camera pose, yielding RGB images $\mathbf{I}^{t,v}$ and disocclusion masks $\mathbf{M}^{t,v}$ through $(\mathbf{I}^{t,v}, \mathbf{M}^{t,v}) = \mathcal{R}(\mathbf{P}^{t,v}, \mathbf{C}^t)$, where the rendering function $\mathcal{R}$ projects the 3D point cloud $\mathbf{P}^{t,v}$ onto the 2D image plane according to the camera pose $\mathbf{C}^t$ at time t . Given a sequence of camera poses $\mathbf{C} = (\mathbf{C}^1, \dots, \mathbf{C}^L)$, rendering every cache element produces V videos of length L , which stack into per-view image sequences $\mathbf{I}^v \in \mathbb{R}^{L \times 3 \times H \times W}$ and mask sequences $\mathbf{M}^v \in \mathbb{R}^{L \times 1 \times H \times W}$. The disocclusion masks delineate precisely the regions that the video diffusion model must inpaint, and together the renderings constitute structured visual guidance that conditions the subsequent video generation.

Video variational autoencoder (VAE). Diffusion-based video generators conventionally rely on a video variational autoencoder (VAE) (Kingma & Welling, 2013; Rombach et al., 2022) to compress videos into a lower-dimensional latent space, rendering both training and inference tractable. Given an RGB video $\mathbf{I} \in \mathbb{R}^{L \times 3 \times H \times W}$, a pre-trained VAE encoder $\mathcal{E}$ maps it into this latent space as $\mathbf{z} = \mathcal{E}(\mathbf{I}) \in \mathbb{R}^{L' \times C \times h \times w}$, within which the diffusion model is trained and sampled. The generated video $\hat{\mathbf{I}} = \mathcal{D}_{rgb}(\mathbf{z})$ is then recovered by a pre-trained VAE decoder $\mathcal{D}_{rgb}$. We adopt the pre-trained GEN3C (Ren et al., 2025; Cosmos, 2025) model, for which the latent channel dimension is $C = 16$, the temporal dimension is $L' = (L - 1)/\tau + 1$, and the spatial dimensions are $h = H/\sigma$ and $w = W/\sigma$, with temporal compression factor $\tau = 8$ and spatial compression factor $\sigma = 8$.

3.2 SELF-DISTILLATION

We train the 3DGS decoder $\mathcal{D}_s$ as the student in a teacher–student paradigm whose teacher is a camera-controlled video diffusion model $\mathcal{V}$. To secure diverse supervision, we curate a large-scale

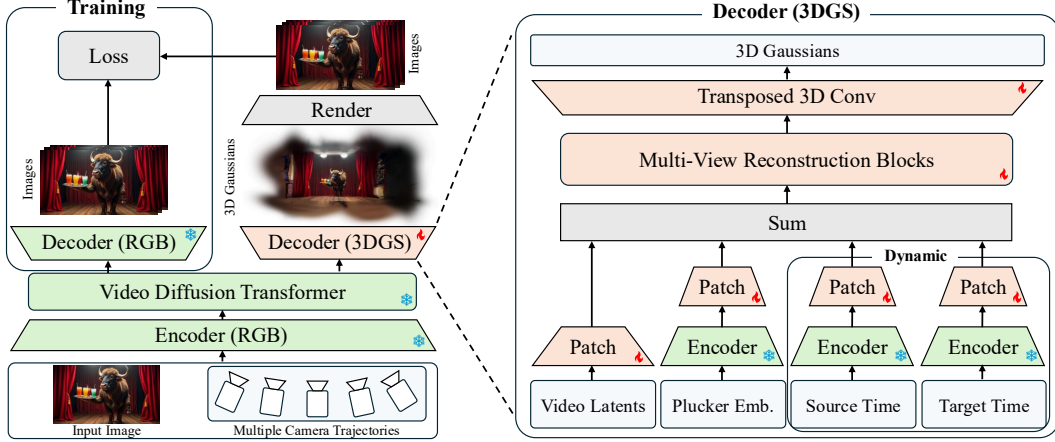


Figure 4: **3D Generative Reconstruction Framework.** Our pipeline builds upon a camera-controlled video diffusion model (Ren et al., 2025) pre-trained on large-scale data. We train a 3D Gaussian Splatting (3DGS) decoder by aligning the 2D renderings of the generated 3DGS scenes with the RGB-decoded outputs of the pre-trained video model, training only the 3DGS decoder while keeping the pre-trained autoencoder and diffusion model frozen. At inference time the 3DGS decoder is used directly, and the RGB decoder is no longer required. Time conditioning within the 3DGS decoder further lets us extend the approach seamlessly from static to dynamic 3D scene generation.

collection of text prompts with large language models (OpenAI, 2025; Bai et al., 2023), synthesize images I with an image diffusion model (Black Forest Labs, 2024), and expand each image into multi-view sequences using GEN3C (Ren et al., 2025).

Teacher-student setup. Given an input image I and a sampled camera trajectory $\{\mathbf{C}^t\}_{t=1}^L$, the video diffusion model $\mathcal{V}$ produces a denoised video latent $\mathbf{z} = \mathcal{V}(I, \{\mathbf{C}^t\}_{t=1}^L)$. This latent is decoded along two branches: the pre-trained RGB decoder $\mathcal{D}_{rgb}$ yields video frames $\mathbf{I}_{\mathcal{D}_{rgb}} = \mathcal{D}_{rgb}(\mathbf{z})$, while the 3DGS decoder $\mathcal{D}_s$ emits explicit 3D Gaussians $\mathbf{G}$. The views rendered from the Gaussians (*student*), $\mathbf{I}_{\mathcal{D}_s} = \text{Render}(\mathbf{G}, \{\mathbf{C}^t\}_{t=1}^L)$, are supervised to match the RGB frames $\mathbf{I}_{\mathcal{D}_{rgb}}$ (*teacher*), thereby closing the self-distillation loop.

Multi-trajectory supervision. To enlarge the viewpoint coverage afforded by a single input image, we sample $V = 6$ camera trajectories per image, as shown in Fig. 3. For each trajectory v we construct a spatiotemporal cache $\{\mathbf{P}^{t,v}\}_{t=1}^L$ together with $L = 121$ camera poses $\{\mathbf{C}^{t,v}\}_{t=1}^L$. Passing these caches through the video diffusion model $\mathcal{V}$ yields latents $\mathbf{z}^v$, which the teacher decodes into RGB frames $\mathbf{I}_{\mathcal{D}_{rgb}}^v = \mathcal{D}_{rgb}(\mathbf{z}^v)$. The 3DGS decoder $\mathcal{D}_s$ then learns to fuse the multiple $\mathbf{z}^v$ into a single coherent set of Gaussians $\mathbf{G}$ while inpainting disoccluded regions. This design trains $\mathcal{D}_s$ entirely from synthetic supervision provided by $\mathcal{V}$: operating in latent space permits efficient aggregation across trajectories, whereas the explicit Gaussian output $\mathbf{G}$ enforces geometric consistency and directly supports downstream simulation and real-time rendering.

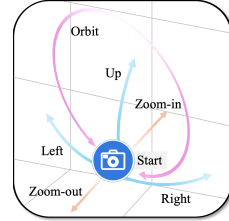


Figure 3: The six sampled camera trajectories that jointly maximize view coverage.

4 FEED-FORWARD RECONSTRUCTION FROM MULTI-VIEW VIDEO LATENTS

Our feed-forward 3DGS decoder $\mathcal{D}_s$ transforms the multi-view latents synthesized by the video model $\mathcal{V}$ into an explicit 3D representation renderable from arbitrary viewpoints. We adopt 3D Gaussian Splatting (3DGS) (Kerbl et al., 2023) for its explicit parameterization and fast rendering, and detail the decoder architecture in Fig. 4 (right).

4.1 3DGS DECODER

Scaling Latent-Based 3D Reconstruction. Prior feed-forward reconstruction frameworks generate one 3D Gaussian per input pixel and are therefore bounded by the number and resolution of views they

can ingest: GS-LRM (Zhang et al., 2024e) operates on 2–4 images at 512×512 , and AnySplat (Jiang et al., 2025a) trains on 24 images at 448×448 . By contrast, our video model synthesizes $V \times L = 6 \times 121 = 726$ input views at 704×1280 resolution, a regime that far exceeds the capacity of existing methods and overruns available GPU memory. The bottleneck is the attention over visual tokens, whose memory and compute scale with the pixel count. We circumvent it by declining to scale in pixel space at all, and instead operate directly in the compressed video latent space produced by our camera-controlled video diffusion model. We denote the resulting multi-view video latents as $\mathbf{Z} \in \mathbb{R}^{V \times L' \times C \times h \times w}$.

Architecture. The 3DGS decoder $\mathcal{D}_s$ maps its inputs into the hidden dimension of the main reconstruction blocks and predicts per-pixel 3D Gaussian features $\mathbf{G} \in \mathbb{R}^{V \times L \times H \times W \times 14}$. Its inputs are the multi-view video latents $\mathbf{Z}$ and the encoded Plücker embeddings $\mathbf{E}$, both of latent dimension $\mathbf{Z}, \mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. Each input is patchified to the block hidden dimension by a single spatial 2×2 patchification layer (Dosovitskiy et al., 2021); notably, no additional visual encoder is required, since the video latents already provide a compact token representation. The two patchified streams are summed and passed through the reconstruction blocks. Following the design of Long-LRM (Ziwen et al., 2024), each block comprises one Transformer (Vaswani et al., 2017) layer followed by seven Mamba-2 (Dao & Gu, 2024) layers, and we stack the block twice to obtain 16 layers with a 512-dimensional hidden state. A final transposed 3D convolution maps the hidden representation to the 14 Gaussian channels: 3D position (x, y, z) , scale (s_x, s_y, s_z) , rotation quaternion (q_w, q_x, q_y, q_z) , opacity α , and RGB (r, g, b) . The static decoder is thus expressed compactly as $\mathbf{G} = \mathcal{D}_s(\mathbf{Z}, \mathbf{E})$.

Plücker embeddings. We first compute raw Plücker embeddings from the camera poses $\{\mathbf{C}^{t,v}\}$ and intrinsics as $\mathbf{E}^{\text{raw}} \in \mathbb{R}^{V \times L \times 6 \times H \times W}$, and encode them by reusing the RGB encoder $\mathcal{E}$ of the pre-trained VAE. Since $\mathcal{E}$ accepts three-channel input, we encode the 3-dimensional ray directions and the 3-dimensional cross product of ray directions and origins separately, then concatenate the two latents along the channel dimension to obtain $\mathbf{E}^{\text{enc}} \in \mathbb{R}^{V \times L' \times 2C \times h \times w}$. A lightweight MLP finally maps $\mathbf{E}^{\text{enc}}$ to $\mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$, halving the channel dimension so that it aligns with the video latent $\mathbf{Z}$.

4.2 LOSS FUNCTION

Image-based supervision. We supervise the 3DGS decoder with an image-based reconstruction objective that decomposes into a Mean Squared Error (MSE) term $\mathcal{L}_{mse}$ and a perceptual LPIPS term $\mathcal{L}_{lpips}$ (Zhang et al., 2018), the latter computed with VGG (Simonyan & Zisserman, 2014) as the feature extractor.

Depth supervision. A decoder trained under RGB supervision alone tends to collapse geometry into flattened surfaces. To counteract this, we additionally supervise the rendered depth maps against the temporally consistent video depth estimated by the off-the-shelf system ViPE (Huang et al., 2025), using the scale-invariant depth loss $\mathcal{L}_{depth}$ of Long-LRM (Ziwen et al., 2024).

Opacity-based pruning. Following Long-LRM (Ziwen et al., 2024), we impose an L_1 regularizer on opacity, $\mathcal{L}_{opacity}$, and prune the Gaussians occupying the lowest 80% of opacity.

Total loss. The total objective is the weighted sum of the individual terms, with weighting factors λ_i ,

$$\mathcal{L} = \lambda_{mse} \mathcal{L}_{mse} + \lambda_{lpips} \mathcal{L}_{lpips} + \lambda_{depth} \mathcal{L}_{depth} + \lambda_{opacity} \mathcal{L}_{opacity} \quad (1)$$

where we set $\lambda_{mse} = 1.0$, $\lambda_{lpips} = 0.5$, $\lambda_{depth} = 0.05$, and $\lambda_{opacity} = 0.1$ throughout.

5 EXTENSION TO DYNAMIC 3D SCENES

Our framework extends to dynamic 3D scenes with only minimal changes, which we detail below.

Self-distillation for dynamic scenes. The dynamic setup mirrors its static counterpart closely. In place of a single image, the video model now ingests a single video of length L together with its camera poses and generates multi-view video latents that capture the same underlying motion. For video inputs we follow a protocol analogous to the static case: we sample diverse text prompts with large language models and then generate videos with video diffusion models (Cosmos, 2025; Wan, 2025), subsequently annotating each generated video with camera poses and depth maps via ViPE (Huang et al., 2025).

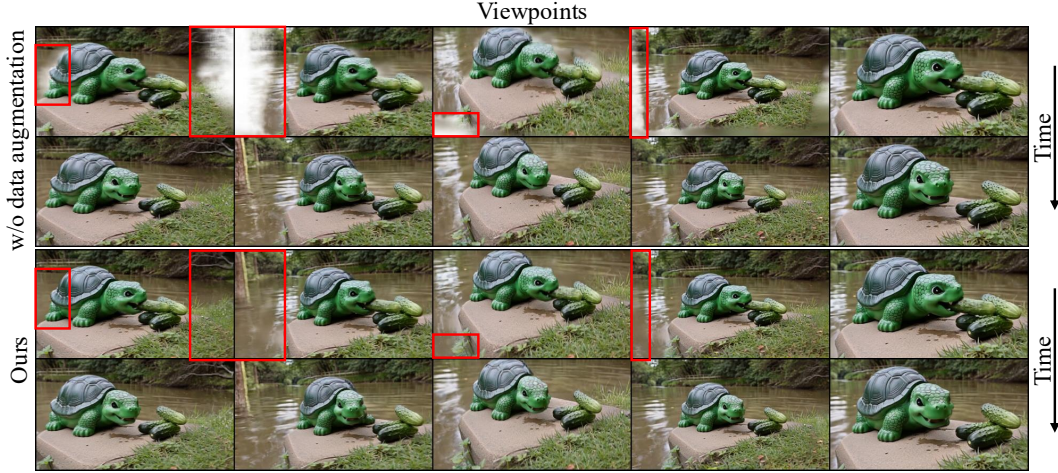


Figure 5: **Dynamic data augmentation.** Naively supervising the time-aware 3DGS decoder induces artifacts in the generated 3D Gaussians: at early timesteps, regions uncovered by the supervision signal collapse to low opacity under extreme poses. Augmenting the supervision with a motion-reversed video ensures that every timestep is observed with full spatial coverage, thereby eliminating these low-opacity artifacts at early timesteps.

Dynamic 3DGS decoder architecture. Adopting the bullet-time formulation of Liang et al. (2025b), we design the dynamic 3DGS decoder $\mathcal{D}_s$ to predict time-dependent 3D Gaussians for every video frame. The decoder $\mathcal{D}_s$ retains the architecture of $\mathcal{D}_s$ and augments its input with encoded source and target time embeddings $\mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The raw source times are assigned per input frame as $\mathbf{T}^{\text{src, raw}} \in \mathbb{R}^{V \times L \times 1 \times H \times W}$, and the raw target time as $\mathbf{T}^{\text{tgt, raw}} \in \mathbb{R}^{V \times 1 \times 1 \times H \times W}$. Both are spatially replicated and combined with a 2-dimensional sinusoidal embedding to expand the channel dimension to 3, then encoded by the RGB encoder $\mathcal{E}$ into latents $\mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}} \in \mathbb{R}^{V \times L' \times C \times h \times w}$, with the target time replicated along the latent temporal dimension to match L' . At training time we sample a target timestep at random, decode the corresponding 3D Gaussians, and supervise them across all trajectories produced by dynamic data augmentation. We fine-tune $\mathcal{D}_d$ from the pre-trained $\mathcal{D}_s$, initializing the patchification layers of $\mathbf{T}^{\text{src}}$ and $\mathbf{T}^{\text{tgt}}$ to zero; $\mathbf{T}^{\text{src}}$ and $\mathbf{T}^{\text{tgt}}$ are then added directly to the sum of $\mathbf{Z}$ and $\mathbf{E}$. The dynamic decoder is thus defined as $\mathbf{G} = \mathcal{D}_d(\mathbf{Z}, \mathbf{E}, \mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}})$.

Dynamic data augmentation. In static scenes, where time plays no role, frames from every timestep are jointly valid supervision for the 3DGS renderings. In dynamic scenes, by contrast, the 3DGS evolves over time, so only frames from the matching timestamp constitute valid supervision. This constraint admits a degenerate solution in which the 3DGS at a given timestep simply disregards information from frames at other timesteps (see Fig. 5).

We resolve this failure mode with a dynamic data augmentation strategy that balances supervision across timesteps. Early frames are naturally tied to viewpoints near the input image, whereas later frames correspond to more distant viewpoints; to redress this imbalance, we furnish each timestep with paired supervision views, one near and one far. Concretely, we reverse the input video in frame order and feed it to the video model $\mathcal{V}$, obtaining six additional multi-view sequences. Reversing these sequences back to the original motion order yields six inward-progressing trajectories (from distant viewpoints toward the input) that, combined with the original six outward trajectories, provide 12 supervision views per timestep. This augmentation is confined to training: the reversed trajectories serve purely as supplementary supervision that forestalls collapse from pruning artifacts, and are not needed at inference.

6 EXPERIMENTS

6.1 EXPERIMENTAL SETUP

Datasets. We train our model without recourse to any existing multi-view dataset; instead, we construct our own, the *Lyra* dataset, supervising the 3DGS decoder entirely from our video model. The 3D reconstruction setup comprises 59,031 images and the 4D setup 7,378 videos, all originating

Table 1: **State-of-the-art comparisons.** Single-image-to-3D generation on RealEstate10K, DL3DV, and Tanks-and-Temples. Our method attains the best PSNR, SSIM, and LPIPS on every dataset, surpassing all prior methods across all metrics; the best value in each column is shown in **bold**.

Method	RealEstate10K			DL3DV			Tanks-and-Temples		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
ZeroNVS	13.01	0.378	0.448	13.35	0.339	0.465	12.94	0.325	0.470
ViewCrafter	16.84	0.514	0.341	15.53	0.525	0.352	14.93	0.483	0.384
Wonderland	17.15	0.550	0.292	16.64	0.574	0.325	15.90	0.510	0.344
Bolt3D	21.54	0.747	0.234	-	-	-	-	-	-
Ours	21.79	0.752	0.219	20.09	0.583	0.313	19.24	0.570	0.336



Figure 6: **Image-to-3DGS Generation.** Five novel views rendered from our generated 3DGS scenes.

from diverse text prompts that span indoor and outdoor environments, humans, animals, and both realistic and imaginative content. Synthesizing six camera trajectories per image (3D) or video (4D) yields 354,186 videos for the 3D setup and 44,268 videos for the 4D setup.

Baselines. We benchmark our method against the state-of-the-art approaches for generative single-image-to-3D reconstruction: ZeroNVS (Sargent et al., 2024), ViewCrafter (Yu et al., 2024b), Wonderland (Liang et al., 2025a), and Bolt3D (Szymanowicz et al., 2025b). Because *no source code is available* for these methods, we cannot re-evaluate them on our out-of-distribution set and therefore rely on the quantitative results reported in the respective papers.

Evaluation. Consistent with prior work, we evaluate single-image-to-3D on RealEstate10K (Zhou et al., 2018), DL3DV (Ling et al., 2024b), and Tanks-and-Temples (Knapitsch et al., 2017), adopting the evaluation protocol of Wonderland (Liang et al., 2025a) and Bolt3D (Szymanowicz et al., 2025b) and reporting the standard reconstruction metrics PSNR, SSIM, and LPIPS.

6.2 MAIN RESULTS

Quantitative results. Tab. 1 reports the single-image-to-3D evaluation. Our method surpasses all prior work on every benchmark and across every metric, i.e., higher PSNR and SSIM and lower LPIPS on RealEstate10K, DL3DV, and Tanks-and-Temples alike. Because the reconstruction is distilled from the video model, this result also implies a clear path to further gains: stronger video generative backbones will translate directly into improved reconstructions. Additional quantitative comparisons on the *Lyra* dataset are provided in Appendix C.

Qualitative results. Fig. 1 and Fig. 6 present novel-view renderings from our generated 3DGS scenes; video results will be released upon acceptance. Our method synthesizes high-quality novel-view content in previously unobserved regions of 3D and 4D scenes while remaining faithful to the input image or video. Qualitative comparisons on the *Lyra* dataset appear in Appendix C.



Figure 7: **Ablations.** Each ablation is visualized by rendering the same extreme novel viewpoint after image-to-3DGS generation; depth visualizations and further ablations appear in Appendix C.4.

6.3 ABLATIONS

Tab. 2 substantiates our design choices by ablating each key component on out-of-distribution diverse prompts, evaluating the 3DGS renderings against the videos generated by our camera-controlled video diffusion model.

Real data only. Replacing self-distillation with training on only the real multi-view datasets RealEstate10K (Zhou et al., 2018) and DL3DV (Ling et al., 2024b) — the datasets favored in prior work — sharply degrades generalization to out-of-distribution scenes, from 24.77 to 19.08 PSNR.

Self-distillation + real data. Notably, augmenting self-distillation with real data yields no improvement over self-distillation alone (24.74 vs. 24.77 PSNR). This confirms that the synthetic supervision from self-distillation is already sufficiently diverse and consistent to learn the reconstruction model on its own.

No depth loss. The depth loss prevents flattened geometry and even marginally improves the image-based metrics (24.77 vs. 24.31 PSNR); corresponding depth maps are visualized in Appendix C.4.

No opacity pruning. Retaining all Gaussians rather than pruning by opacity is both less compact and slightly worse in visual quality: learning to prune the majority of the Gaussians yields a more compact representation while improving quality. Moreover, at $H = 704$ and $W = 1280$, rendering takes 18ms with pruning versus 30ms without, a $1.67\times$ speedup.

No LPIPS loss. The LPIPS loss confers robustness to input inconsistencies and better preserves high-frequency detail, and its removal degrades LPIPS markedly (0.224 to 0.370).

No multi-view fusion. A naive alternative generates 3D Gaussians independently per camera trajectory and fuses the resulting point clouds post hoc. Learning the multi-view fusion instead within our reconstruction blocks, where each token attends to all others, is decisively better, raising PSNR from 17.73 to 24.77.

No Mamba-2. Replacing our joint Transformer/Mamba-2 blocks with Transformer-only blocks lowers quality slightly while incurring a substantial efficiency penalty: a single forward pass with $V = 6$, $L = 121$, $H = 704$, and $W = 1280$ takes 3213ms with the joint blocks versus 20922ms with Transformer-only blocks, a $6.5\times$ speedup.

No latent-based 3DGS. Pixel-space methods such as BTimer (Liang et al., 2025b) are confined to 12 input frames, whereas our latent-space formulation accommodates up to 726. Operating the 3DGS decoder in pixel space rather than latent space consequently runs out of memory, underscoring that the latent formulation is what makes this input scale feasible.

Table 2: **Ablation study on the *Lyra* dataset.** Removing any component of our design degrades performance relative to the full model (top row), confirming that each contributes to the final quality.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Ours	24.77	0.837	0.224
Data			
real data only	19.08	0.659	0.413
self-distill. + real data	24.74	0.823	0.236
Loss			
w/o depth loss	24.31	0.811	0.247
w/o opacity pruning	24.55	0.820	0.237
w/o LPIPS loss	23.74	0.766	0.370
Architecture			
w/o multi-view fusion	17.73	0.632	0.446
w/o Mamba-2	24.58	0.818	0.241
w/o latent 3DGS		OOM	

7 CONCLUSION

We have presented *Lyra*, a unified 3D and 4D generation framework driven by a single image or video. Rather than collecting multi-view datasets, we cast camera-controlled video diffusion models as teachers within a self-distillation framework that trains a student 3DGS decoder operating directly in the video model’s latent space and reconstructing 3D Gaussians without any post-processing or optimization. This design broadens generalization and coverage across diverse scenes, and its dynamic extension establishes the feasibility of 4D generation from monocular video. The scale and consistency of our generated scenes are presently bounded by the capacity of the underlying camera-controlled video diffusion model; consequently, advances in that backbone will translate directly into larger-scale, more consistent scene synthesis. Two further directions are especially promising: adapting auto-regressive techniques (Chen et al., 2024a) to our framework for large-scale generation, and modeling motion and tracking within the reconstruction network, as in concurrent work (Lin et al., 2025), to further improve visual motion quality.

8 ETHICS STATEMENT

As with all generative AI technologies, our method carries a risk of misuse, notably the production of misleading or synthetic 3D/4D content. That risk notwithstanding, the central aim of this research is to advance simulation, robotics, and embodied AI by furnishing scalable and controllable tools for data generation. We accordingly anticipate that its most significant contributions will arise through the following capabilities and insights.

- **Realistic simulation for embodied agents:** We provide controllable and physically consistent camera motion within generated 3D and 4D environments, supporting the training and evaluation of agents that must perceive, navigate, and interact with complex scenes.
- **Improved scalability for data engines:** By removing the need for multi-view capture and expensive per-scene optimization, our method produces high-fidelity, customizable scenes at scale, which is particularly well suited to research in robotics, reinforcement learning, and closed-loop simulation.
- **Better technical understanding:** As an academic study, this work contributes a foundational case study of how camera control interacts with spatiotemporal generative models, of direct benefit to research in computer vision, graphics, and embodied AI.

While we caution against deceptive or wrongful use of this technology—where realistic generative outputs could yield misleading or unverifiable media—we hold that the adoption of appropriate safeguards, such as provenance tracking, dataset documentation, and sound evaluation practices, will help ensure that generative models advance science and society in positive ways.

9 REPRODUCIBILITY STATEMENT

To enable full reproduction of our results and to accelerate future research in this area, we will release our training and inference code, model weights, and data for both 3D and 4D generation upon acceptance.

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A ADDITIONAL DETAILS OF VIDEO DIFFUSION MODEL

This section elaborates on the GEN3C (Ren et al., 2025) video diffusion backbone introduced in Sec. 3.1 and details how we further strengthen its 3D consistency.

Conservative mask refinement. A principal limitation of GEN3C’s forward warping is background leakage: occluded object parts in the source view (e.g., the lion body in Fig. 8) become visible from novel viewpoints yet are not flagged in the disocclusion mask, which in turn leaves the foreground incompletely inpainted by the video diffusion model, as shown in Fig. 8 (second and fourth columns).

We address this with a deliberately conservative strategy: because the geometry of occluded regions in $\mathbf{P}^{t,v}$ is unknown, we treat all such regions as occupied and derive refined disocclusion masks accordingly, thereby prompting the video model to reason about potentially visible foreground regions that standard point-based rendering misses.

Concretely, we construct a triangular mesh from the camera-space points $\mathbf{P}_c^{t,v} = \mathbf{C}^t \mathbf{P}^{t,v}$ by connecting spatially adjacent pixels, following Hu et al. (2021):

$$\mathcal{M} = \bigcup_{(u,v)} \{ \triangle(\mathbf{p}_{u,v}, \mathbf{p}_{u+1,v}, \mathbf{p}_{u,v+1}), \triangle(\mathbf{p}_{u+1,v}, \mathbf{p}_{u+1,v+1}, \mathbf{p}_{u,v+1}) \} \quad (2)$$

where $\mathbf{p}_{u,v}$ denotes the 3D point at pixel (u, v) in $\mathbf{P}_c^{t,v}$. This surface mesh acts as a boundary separating visible from invisible regions. For each rendered pixel we then compare the standard point-based depth $\mathbf{D}^{t,v}$ with the mesh-interpolated depth $\mathbf{D}_{\mathcal{M}}^{t,v}$ obtained by ray-surface intersection:

$$\mathbf{M}^{t,v}(u, v) = \begin{cases} 0 & \text{if } \mathbf{D}_{\mathcal{M}}^{t,v}(u, v) < \mathbf{D}^{t,v}(u, v) - \epsilon \\ \mathbf{M}_{\text{orig}}^{t,v}(u, v) & \text{otherwise} \end{cases}, \quad (3)$$

where $\mathbf{M}_{\text{orig}}^{t,v}$ denotes the original disocclusion mask from forward warping and ϵ is a small tolerance. Whenever the mesh surface lies closer than the original points, we conservatively mask the region as a potential foreground disocclusion. This yields markedly more reliable guidance for video generation, particularly under large viewpoint changes, as shown in Fig. 8 (third and fifth columns).

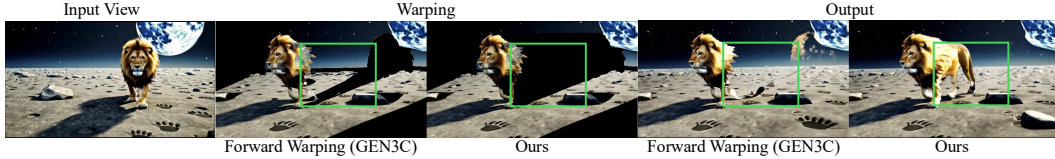


Figure 8: **Rendering function comparison.** Our model builds upon GEN3C (Ren et al., 2025), which conditions camera control on forward-warped images. We contrast forward warping (left) with our conservative rendering function (right): the highlighted region is erroneously filled with background pixels under forward warping but correctly masked by ours, enabling faithful completion.

B MODEL DETAILS

Progressive training. Training the 3DGS decoder directly on 6 trajectories of 121 frames at 704×1280 is prohibitively expensive, so we adopt a progressive training schedule, detailed in Tab. 3, and initialize the dynamic model from the final static pre-trained model. Total training takes 6 days on 8 NVIDIA A100 80 GB GPUs, with gsplat (Ye et al., 2025) as the 3DGS implementation.

Subsampling Gaussians. The count of pixel-aligned 3D Gaussians scales linearly with both spatial and temporal resolution: our generated scenes span 726 frames at 704×1280 , which would entail 654,213,120 per-pixel Gaussians. We therefore emit a single 3D Gaussian per 8×8 spatial neighborhood, reducing the total by a factor of 64 to 10,222,080, and our opacity-based pruning contracts this further to 2,044,416, yielding a substantially more compact representation.

Joint Transformer-Mamba blocks. We adopt the reconstruction block of Long-LRM (Ziwen et al., 2024): one Transformer layer followed by seven Mamba-2 layers, stacked twice to give 16 layers with a 512-dimensional hidden state. Our network is thus considerably smaller than those

Table 3: **Progressive training setup.** Training proceeds in stages that progressively increase the spatial resolution $H \times W$, the temporal length L , and the number of views V , culminating in the dynamic stage.

Stage	$H \times W$	L	V	S	B	Steps
Static						
1	176×320	17	1	17	4	10k
2	176×320	49	1	49	4	2.5k
3	352×640	49	1	49	2	2.5k
4	704×1280	49	1	49	1	2.5k
5	704×1280	121	1	9	1	57.5k
6	704×1280	121	1–6	9	1	7k
Dynamic						
7	704×1280	121	6	12	1	10k

of Long-LRM (Ziwen et al., 2024) and Wonderland (Liang et al., 2025a), which use 24 layers at 1024 hidden dimensions. While pure Transformers are generally more expressive, they are markedly slower in both training and inference than Mamba-2. Since we trained both configurations under an identical budget, we attribute the residual gap to optimization and hypothesize that Transformer-only blocks would require substantially more iterations to match the rendering quality of the joint blocks.

Camera encoding. Following prior work (Kant et al., 2024; Zhang et al., 2024e), we encode cameras with Plücker embeddings. The embeddings are computed at full spatial and temporal resolution and projected into the video latent space by the pre-trained RGB encoder. Because that encoder expects 3 input channels while the Plücker embeddings are 6-dimensional, we encode the 3-dimensional ray directions and the 3-dimensional cross product of ray directions and origins separately, then concatenate the two latent encodings along the channel dimension.

Time encoding. Analogously, to render the 1-dimensional time input compatible with the encoder’s 3-channel interface, we concatenate a 2-dimensional sinusoidal embedding to the raw time value along the channel dimension, replicate the result across the spatial dimensions, and encode it into the video latent space. This procedure lets us encode the source and target times independently within the compressed latent representation.

Dynamic data augmentation. Fig. 9 visualizes the dynamic data augmentation procedure for two example camera trajectories. The procedure constructs pairs of supervision views that observe the scene from the same motion state but from opposite extreme viewpoints.

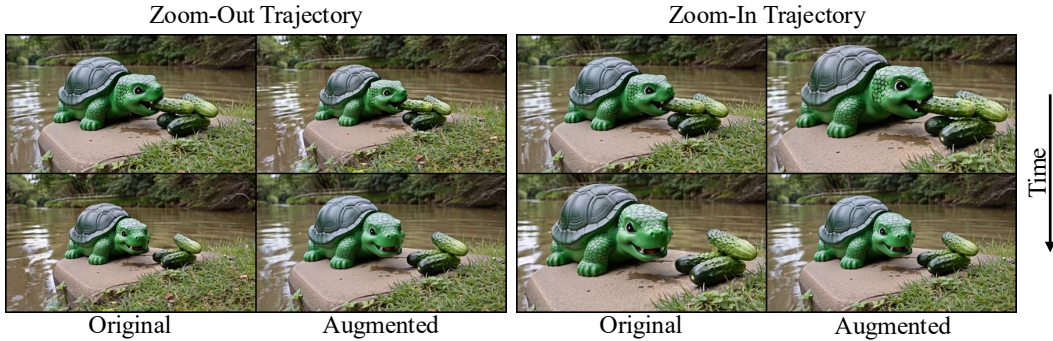


Figure 9: **Dynamic data augmentation videos.** We augment the supervision with a motion-reversed video so that every timestep is observed with full spatial coverage, forestalling low-opacity artifacts at early timesteps. For two example trajectories, zoom-out and zoom-in, we visualize the corresponding augmented videos, whose camera motion is reversed relative to the originals.



Figure 10: **Image-to-3DGS Comparison.** We compare our method with BTimer (GEN3C), showing five views rendered from the generated scenes. Our results exhibit substantially fewer artifacts and higher fidelity.

C ADDITIONAL EVALUATION

C.1 ADDITIONAL BASELINE DETAILS

We additionally compare against BTimer (Liang et al., 2025b), a recent method for joint 3D and 4D reconstruction from posed images or videos. To ensure a fair comparison, we asked the BTimer authors to run experiments on the same scenes as *Lyra*. Since BTimer is purely regression-based, we pair it with our GEN3C (Ren et al., 2025) video diffusion backbone: we use identical camera trajectories and feed the RGB-decoded GEN3C videos as input to BTimer. BTimer operates in pixel space and expects 12 input images; as noted in the main paper, however, pixel-space models such as BTimer exhaust memory when handed the full 726 high-resolution GEN3C frames directly. We therefore uniformly subsample the GEN3C frames: for static evaluations we select 12 frames that evenly span the available viewpoints, and for dynamic evaluations we sample 11 frames across the viewpoint range plus the bullet-time frame of the motion being reconstructed, guaranteeing that the target motion state is always present as a reference. We denote this integrated baseline BTimer (GEN3C), i.e., BTimer combined with GEN3C.

C.2 ADDITIONAL STATIC 3D EVALUATION

For image-to-3DGS generation, Fig. 10 presents qualitative comparisons with BTimer (GEN3C) and Tab. 4a the corresponding quantitative results. Our method attains higher fidelity with markedly fewer artifacts, improving PSNR from 16.32 to 24.92.

C.3 DYNAMIC 3D EVALUATION

BTimer (GEN3C) comparison. For dynamic 3D evaluation, we compare against BTimer (GEN3C) (Sec. C.1) on 100 out-of-distribution videos from our dynamic *Lyra* dataset, cropping each video to 512×512 for a fair comparison, since BTimer was trained predominantly at that resolution. As reported in Tab. 4b, our method substantially outperforms BTimer (GEN3C), raising PSNR from 20.29 to 23.07.

Table 4: **Quantitative results on the static and dynamic *Lyra* datasets.** Across both settings, our method consistently surpasses BTimer (GEN3C) in PSNR, SSIM, and LPIPS; the best value in each column is shown in **bold**.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
BTimer (GEN3C)	16.32	0.580	0.427	BTimer (GEN3C)	20.29	0.687	0.315
Ours	24.92	0.834	0.183	Ours	23.07	0.779	0.231

(a) **Static *Lyra* dataset.** Our method outperforms BTimer (GEN3C) on all metrics.

(b) **Dynamic *Lyra* dataset.** Our method outperforms BTimer (GEN3C) on all metrics.

C.4 DEPTH VISUALIZATION

Fig. 11 visualizes depths recovered from our generated 3DGS. Depth supervision prevents flattened geometry without any sacrifice in visual quality.

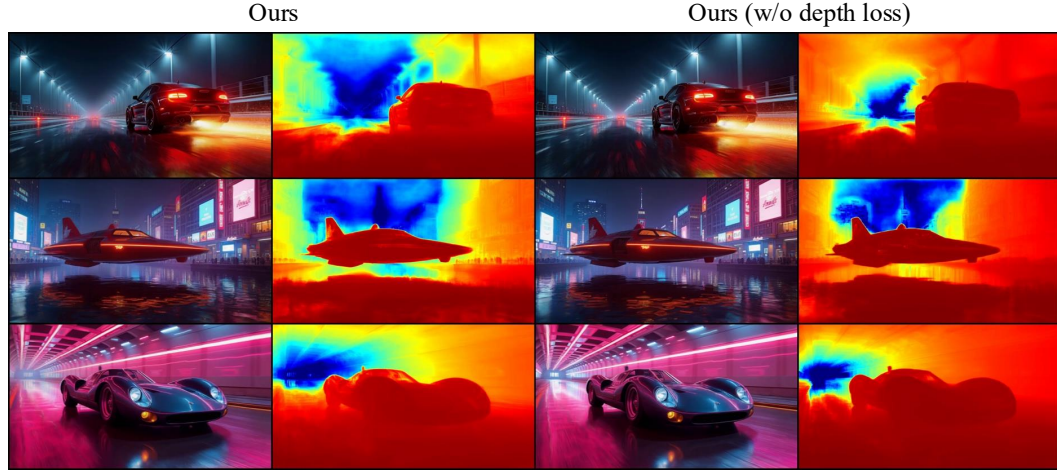


Figure 11: **Depth loss ablation.** 3DGS renderings and their corresponding depth maps. Adding the depth loss as auxiliary supervision prevents flattened geometry without sacrificing visual quality.

D ROBOT SIMULATION

A primary motivation for building a diverse 3D generator is its capacity to synthesize simulation environments for autonomous agents. To this end, we develop an end-to-end pipeline: we generate 3D scenes from text with our model, export the 3D Gaussians as a .ply file, and convert it to .usdz format using 3DGUT (Wu et al., 2025a) and its export function. The resulting .usdz file imports directly into the NVIDIA Isaac robot development platform as a physically based virtual environment, within which AI-based robots can be trained and tested under diverse conditions. Early demo results will be released upon acceptance; a screenshot of the interface is shown in Fig. 12.

E RELATED WORK

This section surveys additional related work and, in Fig. 13, contrasts our approach with the prior methods CAT3D (Gao et al., 2024b) and Bolt3D (Szymanowicz et al., 2025b).

3D generation. Early 3D generation centered on single object categories, extending GANs into 3D by employing neural renderers as an inductive bias (DeVries et al., 2021; Chan et al., 2022; Or-EI et al., 2022; Schwarz et al., 2022; Bahmani et al., 2023b). The advent of CLIP-based supervision (Radford et al., 2021) shifted the field toward more flexible and diverse asset creation, enabling both text-driven generation and editing (Chen et al., 2018; Jain et al., 2022; Sanghi et al., 2022; Jetchev, 2021; Gao et al., 2023; Wang et al., 2022). More recently, diffusion models have sharply improved quality by supplanting CLIP guidance with Score Distillation Sampling (SDS) (Poole et al., 2023; Wang et al.,

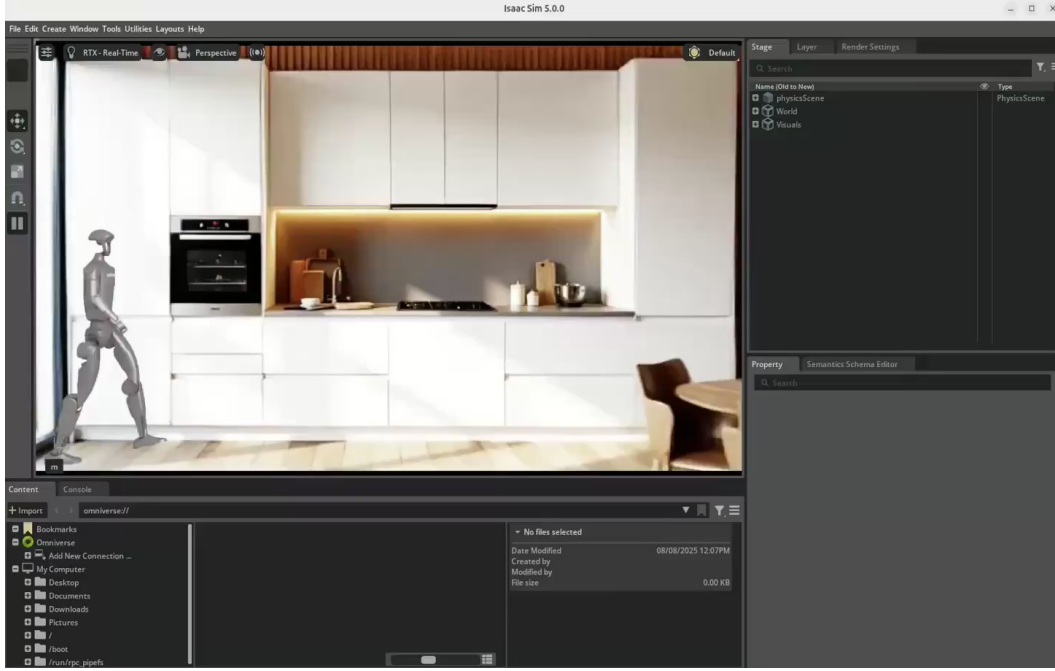


Figure 12: **Robot Simulation.** A frame of a robot simulation running in the Isaac Sim 5.0 framework, using a 3DGS scene generated by our method as the input environment.

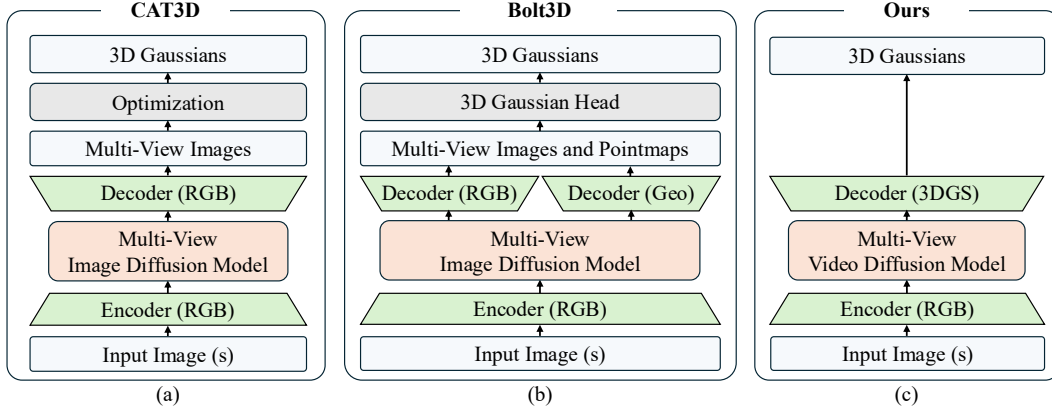


Figure 13: **Approaches for 3D Generation.** CAT3D (Gao et al., 2024b) proposes a multi-view image diffusion model that renders novel viewpoints, which a subsequent optimization stage reconstructs into 3D Gaussians. Bolt3D (Szymanowicz et al., 2025b) fine-tunes CAT3D to emit pointmaps through a geometry autoencoder and, in place of optimization, predicts the 3D scene with a feed-forward 3D Gaussian head. Our work, by contrast, builds upon a pre-trained camera-controlled video diffusion model and *directly* decodes the multi-view video latents into 3D Gaussians.

2023b; Lin et al., 2023a; Chen et al., 2023b; Liang et al., 2023; Wang et al., 2023a; Li et al., 2024e; He et al., 2024b; Ye et al., 2024; Liu et al., 2024b; Yu et al., 2023; Katzir et al., 2024; Lee et al., 2024a; Sun et al., 2024a).

To reinforce structural coherence, several methods enforce multi-view consistency by generating scenes from distinct perspectives (Lin et al., 2023b; Liu et al., 2023; Shi et al., 2024; Feng et al., 2024a; Liu et al., 2024a; Kim et al., 2023; Voleti et al., 2024; Höllein et al., 2024; Tang et al., 2024c; Gao et al., 2024b; Wang et al., 2025f; Kant et al., 2025; Yuan et al., 2025; Ren et al., 2024b), whereas others cast scene synthesis as iterative inpainting (Höllein et al., 2023; Shriram et al., 2024). A further line of work lifts 2D images into 3D using NeRF (Mildenhall et al., 2020), 3D Gaussian Splatting (Kerbl et al., 2023), or mesh-based representations coupled with diffusion models (Chan

et al., 2023; Tang et al., 2023a; Gu et al., 2023; Liu et al., 2024c; Yoo et al., 2023; Tewari et al., 2023; Qian et al., 2024b; Long et al., 2024; Wan et al., 2024; Szymanowicz et al., 2023; Lu et al., 2024).

Feed-forward 3D models. A complementary body of work develops fast, feed-forward techniques that predict 3D content directly from images or text (Hong et al., 2024; Li et al., 2024b; Xu et al., 2024e;d; Zhang et al., 2024a; Han et al., 2024; Jiang & Wang, 2024; Xie et al., 2024a; Tang et al., 2024b; Tochilkin et al., 2024; Qian et al., 2024a; Szymanowicz et al., 2024; 2025a; Liang et al., 2025a; Szymanowicz et al., 2025b; Schwarz et al., 2025; Yang et al., 2025; Zhang et al., 2025a). Unlike our method, however, these approaches remain confined to static 3D scenes, and some target narrow domains such as faces (Kirschstein et al., 2025). Concurrent work (Liang et al., 2025b; Xu et al., 2025) addresses real scenes but cannot accommodate diverse generated scenes or large viewpoint changes.

4D generation. Progress in 4D generation, i.e., dynamic 3D scene synthesis, has been rapid, with many approaches guided by input text prompts or images. Following the introduction of large-scale generative models for the task (Singer et al., 2023), subsequent work has advanced both visual fidelity and motion quality (Ren et al., 2023; Ling et al., 2024a; Bahmani et al., 2024b; Zheng et al., 2024b; Gao et al., 2024a; Yang et al., 2024a; Jiang et al., 2024b; Miao et al., 2024; Li et al., 2024a; Zhang et al., 2024g; Yuan et al., 2024; Jiang et al., 2024a).

Beyond text conditioning, an alternative family of methods lifts 2D images or videos into dynamic 3D scenes (Ren et al., 2023; Zhao et al., 2023; Yin et al., 2023; Pan et al., 2024; Zheng et al., 2024b; Ling et al., 2024a; Gao et al., 2024a; Zeng et al., 2024b; Chu et al., 2024; Wu et al., 2024; Yang et al., 2024b; Wang et al., 2024c; Feng et al., 2024b; Sun et al., 2024c; Zhang et al., 2024c; Ren et al., 2024a; Lee et al., 2024b; Li et al., 2024c; Van Hoorick et al., 2024; Uzolas et al., 2024; Chai et al., 2024; Liang et al., 2024; Li et al., 2024d; Xie et al., 2024b; Li et al., 2025a; Wang et al., 2025e; Zhu et al., 2025; Zhou et al., 2025; Hu et al., 2025; Park et al., 2025). Further efforts inject physics priors into the generation pipeline (Lin et al., 2024; Huang et al., 2024b; Zhang et al., 2024f; Kiray et al., 2025) or sharpen motion controllability through template-driven approaches (Zhang et al., 2024d; Sun et al., 2024b; Chen et al., 2024b), while a parallel direction pursues compositional and interactive 4D generation (Bahmani et al., 2024a; Xu et al., 2024b; Cao et al., 2024; Yu et al., 2024a; Zeng et al., 2024a; Zhu et al., 2024).

A separate strand extends 3D GAN frameworks into the 4D setting by training directly on 2D video (Bahmani et al., 2023a; Xu et al., 2023). Such methods, however, are typically constrained by small-scale, single-category datasets that limit generalization, and most remain object-centric, neglecting background. Their visual fidelity consequently trails the photorealism of recent video generation models, precisely the source of supervision our method exploits.

Camera-conditioned video models. A substantial body of recent work incorporates camera control into video diffusion models. The pioneering MotionCtrl (Wang et al., 2024e) introduced camera conditioning by augmenting pre-trained video models (Chen et al., 2023a; Blattmann et al., 2023) with extrinsic matrices. Subsequent efforts (He et al., 2024a; Xu et al., 2024c; Kuang et al., 2024; Ju et al., 2025; He et al., 2025; Li et al., 2025c; Wang et al., 2025c; 2024d; Lei et al., 2025; Liu et al., 2025) strengthened this conditioning through Plücker-coordinate camera representations, while others follow MotionCtrl in simply flattening the poses (Wang et al., 2025b). A further line (Hu et al., 2024; Xiao et al., 2024; Ling et al., 2024c; Hou et al., 2024) achieves camera-motion control without introducing additional trainable parameters, and yet others adopt low-rank formulations for camera control (Zhang et al., 2025b). All of these, however, rest on U-Net-based architectures.

Several recent methods continue to extend U-Net-based camera control (Zhao et al., 2024; Xu et al., 2024a; Zheng et al., 2024a; Zhang et al., 2024b; Yu et al., 2024b; Lei et al., 2024), while others begin to explore diffusion transformers. Cheong et al. (Cheong et al., 2024) propose one such transformer-based framework, though its scene and visual quality remain limited, and DimensionX (Sun et al., 2024d) introduces joint space-time control in diffusion transformers but relies on pre-defined, non-continuous camera trajectories. Others instead pursue pose and geometry estimation with a video DiT (Huang et al., 2024a; Mai et al., 2025; Jiang et al., 2025b). CAT4D (Wu et al., 2025b) derives a multi-view video diffusion model by fine-tuning a multi-view backbone, and large-scale synthetic data further boosts performance, as SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a) demonstrate with transformer-based architectures. Finally, 4DiM (Watson et al., 2024) trains a space-time diffusion model from scratch for single-image novel-view synthesis. A related direction

1512 edits the camera movement of videos with underlying motion via camera retargeting (Seo et al., 2025;
1513 Ma et al., 2025), and a particularly influential one conditions camera-controlled video generation on
1514 a spatiotemporal cache built from point clouds or optical flow (Wu et al., 2025c; Ren et al., 2025;
1515 Wang et al., 2025d; Cao et al., 2025; YU et al., 2025; Jin et al., 2025; Xing et al., 2025; Yan et al.,
1516 2025; Li et al., 2025b; Gu et al., 2025).